- The advantage of a formal system is that a characterization of the axioms and rules of inference implicitly defines the set of theorems.
- In such a system it becomes possible to distinguish between assertion which are true and those which are provable. An assertion is provable if it is a theorem. We have seen that some assertions are true in some domain but not in others.

Two questions can be asked about a formal system

1. Is it powerful enough to prove all valid assertions, regardless of domain or interpretation.
2. Is it powerful enough to prove all assertions which are true of some particular universe with a specified interpretation of predicate symbols.

- An example here would be the set of integers with predicates corresponding to arithmetic identities. Mathematics has been somewhat successful with question 1 but not with question 2.

Godel's Incompleteness Theorem

- A formal system is consistent if both an assertion A and its negation $\neg A$ cannot be proved in that. It is complete if every true assertion can be proved in the system.

Godel's incompleteness theorem can be stated as follows; There does not exist a complete and consistent system for integer arithmetic.

We try to give a sketch of proof of the above theorem. For this, we require the idea of Godel numbering.

- The Godel number of the finite sequence of positive integers $i_1, i_2, i_3, \ldots, i_n$ is $2^{i_1} * 3^{i_2} * \ldots (\text{Prime}(n-1))^{i_n}$, taking 2 as Prime(0).
- The Godel number of the sequence 3,2,1,1 is $2^3 * 3^2 * 5^1 * 7^1 = 2520$.